

Artificial Intelligence for Corporate Finance

AI-Augmented Excel I: Reading and Improving a Financial Model

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Structure of the course

- **Email:** amedeo.andriollo@dauphine.psl.eu
- **Assessment:** 150-minute final in-class examination.
- **Office hours:** feel free to (DO) come: B612.
- Slides/materials are updated regularly. Report mistakes: corrections help the whole class.
- **This session:** the financial model as a **reviewable system**:
 - ↪ Excel formula + model risk with AI-assisted inspection, diagnosis and *controlled* editing.
- **Built on:** Rosenbaum & Pearl, *Investment Banking*; Koller, Goedhart & Wessels, *Valuation*.

Session 2: the historical workbook arrives

Professional question

Before our team forecasts or values TargetCo, **can we rely** on the historical financial information and analysis in this workbook?

Session	What we receive	What we do
1	Target company profile	Understand the business, identify tensions and formulate diligence questions.
2	Associate-prepared historical workbook	Decode it, screen for candidate issues, verify and classify them, make only approved objective corrections, and document the result.
3	Valuation materials	Build comparables and a simplified DCF; reconcile the valuation.
4	Excel valuation benchmark	Reproduce, test and extend the analysis transparently in Python.

Session question: how can AI help us understand and review an Excel workbook while the analyst retains responsibility for diagnosis, approval and correction?

Deliverables: workbook map; reviewed Audit Assistant and findings report; verified issue classification; A/B EBITDA comparison; approved corrections and 06_Audit_Log.

Refresh (1/3): operating earnings and tax-effected EBIT

Quantity	Economic meaning	TargetCo construction
Revenue	Sales generated by the operating business.	Supplied.
Reported EBITDA	Operating earnings before depreciation and amortisation, interest and taxes.	Supplied.
D&A	Non-cash accounting charges that allocate the cost of capitalised tangible and finite-lived intangible assets over their useful lives.	Supplied.
EBIT	Operating profit after D&A.	$\text{EBIT} = \text{Report.EBITDA} - \text{D\&A.}$
Tax-effected EBIT	Simplified after-tax operating earnings used in this session's cash-flow control, before financing effects.	$\text{Tax-effected EBIT} = \text{EBIT}(1 - t)$ using the stated cash tax rate.

FY2023A example — EUR m

$$62.6 \text{ EBITDA} - 21.4 \text{ D\&A} = 41.2 \text{ EBIT} \implies 41.2(1 - 25\%) = 30.9 \text{ tax-effected EBIT.}$$

Audit implication: a displayed value can be correct today while the model's intended dependency is broken. Numerical agreement alone does not validate the construction.

Refresh (2/3): CapEx and operating working capital

Capital expenditure vs D&A

CapEx are cash outflows during the period for acquiring or improving long-lived operating assets.

D&A is a non-cash accounting expense that allocates capitalised costs over the useful lives of the related assets.

They are economically related over the life of the assets, but they **need not be equal in the same period**.

$$\text{CapEx}_t \neq \text{D\&A}_t$$

Audit implication

errors in the sign, period reference or linkage of CapEx or operating working capital can change cash-flow analysis without changing EBITDA or EBIT.

Operating working capital

For TargetCo:

$$\text{Operating working capital}_t = \text{AR}_t + \text{Inventory}_t - \text{AP}_t.$$

FY2023A, EUR m

Trade receivables 50.0

Inventory 44.1

Trade payables (33.8)

Operating working capital **60.3**

Let OWC denote operating working capital, and:

$$\Delta \text{OWC}_t = \text{OWC}_t - \text{OWC}_{t-1}.$$

FY2022A operating working capital is (56.8), so:
 $\Delta \text{OWC}_{2023A} = 60.3 - 56.8 = +3.5$.

- $\Delta > 0$: decrease in OWC = source of cash.
- $\Delta < 0$: increase in OWC = use of cash.

Refresh (3/3): cash flow and leverage

Simplified free-cash-flow control

$$FCF_t = \text{Tax-effected EBIT}_t + D\&A_t - \text{CapEx}_t - \Delta OWC_t.$$

FY2023A, EUR m

Tax-effected EBIT	30.9
add D&A	21.4
less CapEx	(26.0)
less increase in operating working capital	(3.5)
Simplified FCF control	22.8

Why use this control? It tests whether several upstream calculations are mutually consistent. It is not intended to reconstruct TargetCo's reported historical cash-flow statement.

Debt, cash and leverage

Gross interest-bearing debt: Interest-bearing borrowings outstanding at the measurement date.

Net debt: Gross interest-bearing debt — Eligible unrestricted cash.

Net debt-to-EBITDA: Net debt divided by the specified EBITDA basis for the relevant period.

Critical audit point

A net debt-to-EBITDA ratio is not fully defined until **the EBITDA basis and period** are specified:

i) Reported EBITDA?, ii) Management-adjusted EBITDA?

Both calculations may be arithmetically correct while the evidence remains insufficient to determine which EBITDA basis should be used.

Turn finance definitions into model checks

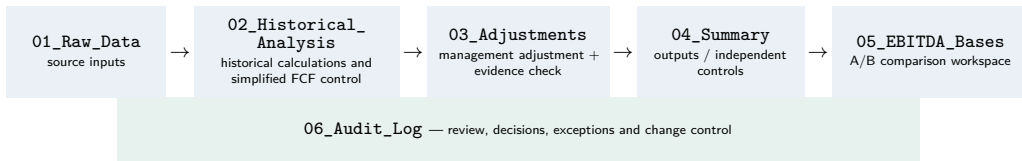
Start from the workbook's stated financial definitions and relationships, then test whether the formulas implement them consistently.

Finance definition / relationship	Independent model check	What the check can reveal
EBIT dependency $\text{EBIT} = \text{Report.EBITDA} - \text{D\&A}$	Recompute EBIT and inspect whether the formula dependency is preserved across periods.	A displayed value may reconcile numerically while the intended dependency is broken.
Operating working capital $\text{Trade receivables} + \text{Inventory} - \text{Trade payables}$	A displayed value may reconcile numerically while the intended dependency is broken.	A construction error can distort both operating working capital and the simplified FCF control.
Net debt and leverage $\text{Net debt} = \text{Gross interest-bearing debt} - \text{Eligible unstr. cash}$ $\text{Net debt} / \text{specified EBITDA basis}$	A construction error can distort both operating working capital and the simplified FCF control.	A ratio can be arithmetically valid while implementing the wrong numerator or EBITDA basis.
Simplified FCF control $\text{EBIT} + \text{D\&A} - \text{CapEx} - \text{delta OWC}$	Recompute the control from its components and trace any difference upstream.	A downstream failure can expose an error in one of several upstream dependencies.
Reported and management-adjusted EBITDA	Preserve and propagate both bases consistently while the evidence remains unresolved.	Arithmetic cannot resolve an evidence-dependent judgment.

Revenue growth and same-period margins remain useful calibration checks, but they do not replace tracing definitions, dependencies and basis consistency.

The workbook is a dependency system

OPEN NOW: TargetCo_historical_model.xlsx



Describe both cell construction and analytical role

Source constant: Supplied value rather than a formula.

Same-sheet formula: Calculation using cells on the same worksheet.

Cross-sheet link: Formula retrieving information from another worksheet.

Blank/ text field: Meaning must be established from context.

Treat sheet names, labels and formatting as cues—not proof. Verify cell role from formulas, dependencies, stated definitions and workbook notes.

Then map relationships

Dependency: Which cells feed which other cells?

Formula family: Which comparable cells appear to implement the same structural rule?

Pattern break: Where does the observed construction differ from the repeated pattern?

Source basis: KGW Ch. 13 sheet discipline.

Reference

A pattern break is a signal to investigate, not a diagnosis

The rule

A structural difference tells us **where to investigate**. By itself, it does not establish the construction is wrong. Three different explanations can produce an unusual cell:

Candidate issue

A calculated row contains formulas in comparable periods except for one constant.

Ask: Was an intended dependency lost?

Candidate exception

A first observable period uses a different construction because the predecessor required by the normal formula is unavailable.

Ask: Does the workbook document and justify the different treatment?

Not yet established

A field is blank because the evidence required to complete the analysis has not been supplied.

Ask: What evidence is required, and is the field intentionally unresolved?

Mind the gap

pattern break $\neq$ confirmed error.

Observe the difference → Establish the expected rule → Check documents → Perform an independent test → **Only then** classify.

The broad "audit and fix" prompt

Prompt

Audit this Excel workbook, find the errors and fix them.

What has the analyst left unspecified?

That single sentence implicitly delegates four important decisions to the AI:

1. Interpret: decide what unusual cells or constructions are supposed to mean;
2. Complete: decide whether blanks or apparently missing values should be populated;
3. Choose: decide whether blanks or apparently missing values should be populated;
4. Act: Select between competing finance treatments when judgment remains open.

↪ the prompt collapses several distinct stages into one instruction:

Interpret → Diagnose → Exercise judgment → Approve → Edit

A plausible audit can start from the wrong model

What the broad prompt leaves AI to infer

1. **Cell meaning:** What each cell is supposed to represent;
2. **Error vs. exception:** Whether an unusual construction is an error, a documented exception or an unresolved item;
3. **Blank:** Whether an empty field should be populated or intentionally remains open;
4. **Finance treatment:** Whether management's proposed EBITDA adjustment should be accepted;
5. **Workbook changes:** Which cells, if any, should be modified.

What the broad prompt never asks AI to establish

1. **Workbook architecture:** Trace formulas and dependencies before inferring purpose;
2. **Source vs. calculation:** Identify declared inputs separately from calculated cells;
3. **Documented exceptions:** Establish whether a different construction is intentional and supported;
4. **Independent finance checks:** Test the workbook against explicitly stated financial definitions;
5. **Evidence and judgment:** Separate management claims from conclusions independently supported by the analyst;
6. **Change control:** Define what may be changed, by whom, and only after what approval.

The problem

A logically consistent audit can still be wrong if the workbook was misinterpreted at the start.

Map the workbook before judging it

1. Start a fresh ChatGPT or Claude conversation. 2. Upload TargetCo_historical_model.xlsx 3. Use the full workbook-mapping prompt from the prompt card. 4. Inspection only — do not request or authorize an edit.

Workbook-mapping prompt, summarized

Map the workbook from formulas and dependencies, not from names or appearance. Distinguish:

- *declared source constants;*
- *same-sheet calculations;*
- *cross-sheet links;*
- *outputs;*
- *blanks.*

Identify formula families, stated definitions, documented exceptions, assumptions and open questions. For each blank, ask whether the workbook supplies the evidence necessary to calculate it.

What we are not deciding yet

Map only.

No correctness conclusion.
No missing-value inference.
No finance-treatment choice.
No edit.

Deliverable

One record for each of the six sheets plus the dependency map.

What a useful workbook map must show

A useful map should establish

1. **Sheet purpose:** Each sheet's purpose supported by its formulas and dependencies;
2. **Cell construction:** Declared source inputs, calculations, cross-sheet links and outputs;
3. **Workbook dependencies:** Upstream and downstream relationships across all six sheets;
4. **Formula families:** Repeated structural patterns and the ranges over which they appear;
5. **Definitions and exceptions:** Workbook-stated financial definitions and documented exceptions;
6. **Assumptions and unresolved questions:** What the AI has inferred and what the workbook does not yet establish.

Two ways a workbook map can fail

Incomplete representation

- A sheet name substitutes for a formula.
- A review/Audit log sheet is omitted.
- Cross-sheet dependencies are not followed through the workbook.

Premature interpretation

- A constant is labelled an error because not a formula.
- A blank evidence field is treated as missing data.
- A row or summary label is accepted as the metric definition without tracing the underlying construction.

A good map describes before it judges. Once the map is verified against the workbook, we have a representation to test (not yet a conclusion).

Correct arithmetic can still rest on an unsupported assumption

Where the unsupported step can enter:

Management identifies an EBITDA adjustment → Management characterizes the cost as non-recurring →
AI treats that characterization as established → Management-adjusted EBITDA becomes the denominator →
Net debt-to-EBITDA is calculated correctly

↪ **The arithmetic can be correct while the treatment remains unsupported.**

What the workbook documents

Management identifies an EBITDA adjustment.

Management supplies the amount of the proposed adjustment.

Management characterizes the item as non-recurring and states when the related activity is expected or claimed to cease.

What still requires analyst judgement

Is the cost separable from ordinary operations?

Has the underlying activity actually ceased?

Is there evidence that the cost will not remain in the expected future operating cost base?

What independent evidence would be required to support the adjustment?

Key distinction

A net debt-to-EBITDA ratio can be arithmetically correct even when the EBITDA basis used in the denominator is not yet supported by the evidence.

Separate description from judgment

The workbook-mapping prompt constraint

Map from formulas and dependencies. For every blank, ask whether the workbook supplies the evidence required to calculate it. Do not infer or fill a missing predecessor. Record documented exceptions separately. Map only. No correctness conclusion. No edit.

What changes when we impose that discipline

1. **Observation:** A construction differs from comparable periods/ A cell contains a constant/ A field is blank/ Management supplies an adjusted EBITDA measure
2. **Do not jump to:** “It is an error” / “Hardcode = error” / “A value is missing” / “Use the adjusted denominator”
3. **First establish:** The expected pattern and whether a documented exception exists / Whether it is a declared source input or a calculation that should preserve a dependency / What information would be required to populate it and whether that information exists / What management is claiming and what evidence would support the analyst’s treatment

What comes later

- **Screen construction:** Identify candidate pattern breaks.
- **Verify and classify:** Apply finance definitions and independent checks.
- **Preserve unresolved judgment:** Keep competing EBITDA bases separate when the evidence does not yet support a choice.
- **Correct only after verification and approval:** Editing follows diagnosis; it does not create the diagnosis.

Decomposing the tasks also limits what AI is allowed to conclude at each step.

Try yourself (1/2): 25 minutes

PART A: Represent the workbook

1. **Build the workbook map.** Prepare one record for each of the six sheets using the format shown on the next slide.
2. **State the finance controls.** Record the financial definitions and relationships the workbook is expected to implement.

PART B: Design the audit method

Write down *how you will later*:

3. **Screen formula patterns** Compare construction across periods and related rows;
4. **Perform independent checks** Specify which important financial quantities you will recompute independently;
5. **Classify candidate findings** Define how you will distinguish an objective error, a documented exception, an unresolved judgment and insufficient evidence;
6. **Prioritise confirmed findings** Define how you will decide which confirmed issues matter most.

Practicing the programming before prompting.

For now

Design the method and do not execute the audit yet. Do not diagnose it. Do not correct it. Do not edit TargetCo.

When each step happens

Now	map + controls + written audit plan
Next	compare
After the break	build and test the Excel Audit Assistant
End	reconcile + 06_Audit_Log

A completed map record

Example: 01_Raw_Data

Sheet	01_Raw_Data
Purpose	Stores the supplied historical source data, management adjustment information, and source / definition notes.
Inputs	Declared source values entered as constants.
Calculations	None (or minimal).
Outputs	Source values and definitions used by downstream analysis sheets.
Depends on	No other sheet.
Feeds	02_Historical_Analysis, 03_Adjustments...
Definitions	Units and fiscal-year convention; operating working-capital definition; net-debt definition; reported EBITDA convention; status of the management adjustment.
Open question	Which supplied items are factual source inputs, and which management claims still require independent support before use in analysis?

Now produce the same for the other five sheets

Record definitions before interpreting

When the workbook explicitly states a definition or convention, record the actual wording and/or exact cell before interpreting it. Only after recording what the workbook states can you compare it with what the formulas actually do.

One rule while you map for the next 25 minutes: **change nothing.**

The pre-edit audit plan

	Step	Question it answers	Action and deliverable	Executed
A	Architecture	<i>How is the workbook organised and linked?</i>	Map each sheet, its role, and its upstream and downstream dependencies. <i>Deliverable:</i> the workbook map.	Now.
B	Finance controls	<i>Which financial definitions and relationships should the workbook implement?</i>	Record each financial control that will later be used to test the workbook. <i>Deliverable:</i> finance-control list.	Now.
C	Formula-pattern scan	<i>Where does construction differ from the expected pattern?</i>	Compare comparable cells across periods or related rows. Record cell, observed construction, expected pattern, reason to investigate. Do not yet call the difference an error.	Later.
D	Independent reconciliation	<i>Can important outputs be reproduced independently from source values and the stated finance definitions?</i>	Recompute selected controls: EBIT, operating working capital, net debt, net debt-to-EBITDA and the simplified free-cash-flow control. <i>Deliverable:</i> workbook value, recomputed value, difference.	Later.
E	Classify and assign status	<i>What has the independent check established, and what remains unresolved?</i>	Record the supported classification and assign status: objective / exception / open judgement / insufficient evidence. <i>Deliverable:</i> category + one-sentence reason.	Later.
F	Prioritise	<i>Which confirmed findings deserve attention first?</i>	Consider current downstream impact and future refresh risk. <i>Deliverable:</i> High / Medium / Low + one-sentence reason.	Later.

A zero reconciliation difference proves only that the numerical agreement under the selected control.

It does not establish that the underlying dependency, basis or judgment is appropriate.

Two routes to the same map

Human-first group

- Excel first: Inspect the workbook manually: read the sheets, trace dependencies and draft the six map records.
- AI second: Use the workbook-mapping prompt to challenge and complete your map.
- Verify: Check any AI additions or corrections against Excel before accepting them.

AI-first group

- AI first: Use the workbook-mapping prompt to generate the initial map.
- Excel second: Verify every returned statement against the workbook, sheet by sheet and dependency by dependency.
- Correct: Amend the map wherever the workbook does not support the AI representation.

Same deliverable

Both groups use: i) same TargetCo workbook; ii) same workbook-mapping prompt; iii) Excel and AI; iv) no Python; v) no edit to TargetCo. The comparison concerns the order of work.

After the 25'

What did the human-first group notice before AI?

What did the AI-first group map faster?

Which AI statements required correction or additional verification?

Which order made verification easier?

The question: Which order leaves the analyst better able to support and defend the final map with workbook evidence?

Try yourself (2/2): 25 minutes

Map + controls + audit plan

Human-first: Excel, then P1

AI-first: P1, then verify in Excel

Same deliverable. Change nothing in TargetCo.

Debrief: compare your maps

Compare the order of work

1. What did human-first reviewers notice before using the workbook-mapping prompt?
2. What did AI-first reviewers map faster?
3. Which AI statement required the most verification in Excel?
4. Which omission or unsupported inference mattered most to your final map?

Compare the maps

1. Which omission or unsupported inference mattered most to your final map?
2. Where does the workbook explicitly state its definitions and conventions?
3. Which differences in construction are explicitly documented?
4. Which blanks represent unresolved evidence requirements rather than values to be calculated?

What we settle now

We compare: i) the quality of the workbook and ii) the order in which human and AI work were performed. We do not resolve candidate findings yet: verification and classification come later, after the Excel Audit Assistant report.

Break — 15'

15 minutes

Keep TargetCo open. We return by building a read-only Excel auditing prototype with AI.

Build the Excel Audit Assistant: obtain the VBA

From zero to hero:

1) Open a **fresh ChatGPT or Claude conversation**. 2) Attach: i) TargetCo_historical_model.xlsx; ii) SESSION2_EXCEL_AUDIT_ASSISTANT_SPEC.md. 3) Paste the request below.

Copy/paste this request

I am a finance student and I do not know VBA. Using the attached TargetCo workbook and functional specification, produce **one complete VBA standard-module implementation** of the Excel Audit Assistant. It must satisfy **every requirement** in the specification. Do not give me pseudocode, partial procedures, placeholders or alternative versions. Use `Public Sub AuditOpenTargetCo()` as the macro I will run. Return, in this order: (1) one complete VBA code block; (2) the name of the macro I must run; (3) a short table stating which workbook the code reads from and which workbook it writes to; (4) exact installation instructions for desktop Excel. TargetCo must never be edited, saved, renamed or closed by the macro. All findings must be written to a new report workbook. Do not choose an EBITDA treatment and do not apply any correction. If any requirement cannot be satisfied, tell me **before** giving the code.

Do not move on until you have

- one **complete VBA code block**;
- the entry macro `AuditOpenTargetCo`;
- no “fill this in yourself” placeholders;
- explicit confirmation that TargetCo is **read only** and output goes to a **new report workbook**.

At this stage you are not judging whether the code is good.

Before running: review the AI-generated VBA

OPTION A — STATIC REVIEW

1. Open a **new AI conversation**.
2. Attach `SESSION2_EXCEL_AUDIT_ASSISTANT_SPEC.md`.
3. Copy the **entire VBA code** produced on the previous slide into the chat.
4. Paste the review request on the right.

Why a new conversation?

The reviewer starts from the instructor specification rather than actually defending the code being produced in the original conversation.

If the verdict is **DO NOT RUN**:

Copy the reviewer's required changes back into the original coding conversation, request one complete revised VBA module, then repeat this review.

Do you understand every line of VBA?

AI is assisting you, but you need to set the standard/criteria for an independent review.

Copy/paste this review request

Act as an independent reviewer of the VBA below. Use the attached functional specification as the required standard. Do not rewrite the code yet. Return a table with four columns:

Requirement | PASS / FAIL / UNCLEAR | Evidence in the VBA | Required action. Check every requirement in the specification. For each finance check, verify the intended formula dependencies, not only the numerical result. Flag checks that could miss a wrong reference because two source cells currently have the same value, and distinguish root-cause errors from downstream effects. In particular, identify every place where the VBA: reads a workbook or cell; writes a value or formula; creates a workbook; saves, closes, renames or deletes anything; uses Shell, system, network or external-file operations; fills Analyst disposition; selects an EBITDA treatment; or applies a correction.

Confirm whether **all writes are restricted to a newly created report workbook** and whether **TargetCo remains read only**.

Finish with exactly one verdict: **SAFE TO TEST ON A DISPOSABLE COPY** or **DO NOT RUN**. If the verdict is **DO NOT RUN**, list the minimum changes required. Do not produce replacement code unless I ask.

Test the VBA in Excel

OPTION B — CONTROLLED EXECUTION

1. In Finder/FileExplorer, make a working **copy** of TargetCo_historical_model.xlsx (same filename). Open only this working copy.
2. In Excel choose File → New Workbook.
3. Save the new workbook as Session2_AuditAssistant_Controller.xlsm.
4. Open the VBA editor: Mac: Tools → Macro → Visual Basic Editor || Windows: Developer → Visual Basic.
5. Choose Insert → Module and paste the VBA.
6. Save the controller workbook.
7. Keep the TargetCo working copy open and run AuditOpenTargetCo.

Expected mechanical result

A **new workbook** containing an Audit Report appears. The TargetCo working copy remains open.

Important

At this stage you are checking whether the Assistant **observes and reports the workbook faithfully**.

CHECK THE REPORT AGAINST EXCEL

For at least three (report) rows:

1. Read the reported **Worksheet** and **Cell**.
2. Go to that exact cell in TargetCo.
3. Compare the report's **Formula or value** with the Excel formula bar.
4. Inspect the comparable cells in the same row / neighbouring periods.
5. Ask: Does the report point to the correct cell? | Does it reproduce the cell correctly? | Is the stated reason for attention visible in Excel?

Record only

Check	Yes	No
Report points to the stated Excel cell	☐	☐
Formula/value agrees with Excel	☐	☐
Reason for flag is observable	☐	☐
Analyst disposition remains blank	☐	☐

Run and test the report

FIRST: CREATE ONE KNOWN TEST

Use **only** the **disposable** TargetCo working **copy**.

1. Go to 02_Historical_Analysis!E12.
2. Confirm that the formula bar shows: =E11/E6.
3. Note the displayed value.
4. Replace the formula with the **same displayed value**: 9.2%.
5. Do **not** change any other cell.
6. Run AuditOpenTargetCo again.

What did we just do?

The number still looks correct, but the formula dependency has been deliberately removed. We therefore know the expected audit outcome *before* looking at the report.

NOW READ THE NEW REPORT

Search the new Audit Report for: 02_Historical_Analysis!E12.

Check	Yes	No
The Assistant flags E12	☐	☐
The report identifies a constant / lost formula pattern	☐	☐
The reported value agrees with Excel	☐	☐

Interpret the result

Flagged: the Assistant detected the known break.

Not flagged: the prototype has missed an in-scope test case. Improve the prompt or code later.

RESET BEFORE CONTINUING

Close the edited TargetCo working copy **without saving it**. Reopen a fresh, unchanged copy before analysing the real Audit Report.

This test answers only: "Can the Assistant detect a known problem?"

It does not tell us whether every candidate in the real report is actually an error.

Detection outcomes

Convention

POSITIVE = the audit method flags the item as requiring analyst attention.

NEGATIVE = the audit method does not flag the item.

	ANALYST	
	Issue	Legitimate
FLAGGED	TP	FP
NOT FLAGGED	FN	TN

TP Flagged and independently confirmed as an in-scope issue.

FP Flagged, but independent review shows the construction is legitimate.

FN Genuine in-scope issue the method missed.

TN Legitimate construction correctly left alone.

The VBA report produces candidate flags: → TP/FP/FN/TN can only be assigned after independent review.

Perturbation tests dependency

Controlled test (disposable TargetCo copy only)

A static review asks: *what is in the cell today?*

A perturbation test asks: **if an upstream input changes, do the expected downstream respond?**

DO THIS

1. Open a fresh disposable copy of TargetCo.
2. Go to 01_Raw_Data!E13 (FY2023A capital expenditure).
3. Change: 26.0 → 27.0
4. Let Excel recalculate.
5. Inspect the cells on the right.
6. When finished, close the TargetCo copy **without saving**.

EXPECTED RESPONSE

Cell	Expected	Why
02_Historical_Analysis!E16	27.0	linked CapEx
02_Historical_Analysis!E17	changes	CapEx / revenue
02_Historical_Analysis!E34	−27.0	CapEx enters the FCF control as a cash outflow
02_Historical_Analysis!E36	21.8	simplified FCF falls by EUR 1.0m
04_Summary!E29	21.8	summary links to the FCF control

A perturbation test does not prove the whole model is correct. It tests whether a specified dependency behaves as expected.

From screening to decision

AUDIT ASSISTANT FLAG



INDEPENDENT CHECK



CLASSIFICATION



NEXT ACTION

The Audit Assistant has finished screening

Status after review	What the independent review establishes	What happens next
OBJECTIVE	The workbook construction conflicts with a stated finance rule or required dependency, confirmed independent.	Eligible for correction only after analyst approval .
EXCEPTION	The unusual construction is supported by the workbook's documented definition, convention or exception.	Leave unchanged; retain the supporting evidence.
OPEN JUDGEMENT	The workbook presents an analytical treatment that cannot be resolved mechanically from the supplied evidence.	Preserve the alternatives; identify the evidence needed.
INSUFFICIENT	The available information does not support either an objective-error diagnosis or a justified exception.	State what is missing; do not force a conclusion.

Three different questions

Screening: should this item receive analyst attention? **Verification:** what does an independent check establish? **Decision:** what action is justified by that evidence?

A candidate becomes an approved correction only after verification and human approval.

Classify the candidates

DO THIS

1. Save the separate Audit Report produced by the VBA.
2. Open a **fresh ChatGPT or Claude conversation**.
3. Attach: i) original TargetCo_historical_model.xlsx; the Audit Report produced by your Assistant.
4. Open SESSION2_PROMPT_CARD.pdf and paste the complete **P2 — Classify the candidates** prompt.
5. Ask for **one row for every candidate** in the Audit Report.

Do not ask AI to fix anything

P2 is a classification step.
TargetCo remains unchanged.

THE P2 OUTPUT MUST CONTAIN

Field	What you need
Candidate	Exact item being reviewed
Classification	What the AI thinks is happening
Reason	Why that classification is proposed
Independent check	How you can verify it yourself
Downstream effect	What other calculations may be affected
Status	Objective / Exception / Open judgement / Insufficient evidence

THEN VERIFY IN EXCEL

For each candidate: i) go to the reported workbook location; ii) perform the proposed independent check; iii) inspect any relevant workbook definition or documented exception; iv) decide whether the proposed classification is supported.

Submission standard

Every Audit Assistant candidate is accounted for. Every classification has an independent check. Every row has one of the four permitted statuses. No workbook edit has been made.

P2 debrief: what was actually wrong?

Objective construction errors

Independent verification identifies **six underlying construction errors**, affecting **nine cells directly**.

Workbook location	What is wrong	Required construction
02_Historical_Analysis!F7	FY2024A revenue growth uses the wrong preceding period.	=F6/E6-1
02_Historical_Analysis!G9	FY2025A reported EBITDA margin uses FY2024A revenue.	=G8/G6
02_Historical_Analysis!F11	EBIT is stored as a constant: the current number reconciles, but the formula dependency is lost.	=F8-F10
02_Historical_Analysis!G24	Operating working capital omits trade payables.	=G21+G22-G23
04_Summary!G24	Net debt adds cash instead of deducting eligible cash.	=G22-G23
04_Summary!D25:G25	Net debt / reported EBITDA uses the wrong EBITDA denominator.	Divide by row 16, not row 18.

Count root causes, not every warning

9 directly incorrect cells, but only 6 underlying construction errors. A downstream warning caused by one of these errors is **not automatically a new root cause**.

Not objective errors

documented structural exceptions; blanks representing unresolved evidence requirements; management's proposed EBITDA adjustment. ↔ The EBITDA adjustment remains an **open finance judgement** until the required evidence is established.

Reconcile your Audit Assistant report with this benchmark: what did it find, what did it miss?

A plausible prompt that decides too early

P3-naive

Based on the workbook and management's adjustment schedule, *which EBITDA measure best represents TargetCo's sustainable operating performance?* *Choose the basis you would use and update the relevant metrics.*

A plausible AI answer

Use the management-adjusted EBITDA basis because:

- management describes the costs as non-recurring;
- the adjustment falls from EUR 2.4m in FY2024A to EUR 1.4m in FY2025A;
- management states that the related activities ended late in FY2025A.

Then update: EBITDA margin; EBIT; net debt-to-EBITDA; the simplified FCF calculation.

What has not been established The supplied evidence does not yet establish whether: i) the costs are separable from ordinary operations; ii) the activities actually ceased; iii) the costs are absent from the expected future cost base; iv) independent evidence supports management's treatment.

What did the prompt delegate?

It asked AI not only to analyse the evidence, but to:

1. decide which EBITDA basis "best represents" performance;
2. choose that basis & update the analysis.

The stronger response

Explain both EBITDA bases, but **do not choose or update** before the analyst has defined the evidence threshold and approval right.

The problem is not that the AI answer is implausible. The problem is that the prompt authorises a conclusion.

EBITDA: keep the bases separate

What the workbook gives us

TargetCo has one **reported historical EBITDA** series and a separate **management-proposed adjustment**. Until the adjustment is independently supported, keep the two bases visible rather than selecting one.

REPORTED EBITDA

FY2024A = 58.2

FY2025A = 68.8

Commissioning / duplicate-QA costs remain included in operating expenses in the periods incurred.

MANAGEMENT-ADJUSTED EBITDA

FY2024A = $58.2 + 2.4 = 60.6$

FY2025A = $68.8 + 1.4 = 70.2$

Management proposes adding back the commissioning / duplicate-QA costs.

Do not collapse the distinction

Management-adjusted EBITDA does not automatically mean **analyst-approved normalized EBITDA**.

At this stage:

Reported EBITDA $\neq$ Management-adjusted EBITDA $\neq$ Analyst-approved normalized EBITDA

Next question: what evidence would be required before the analyst could support the proposed adjustment?

Source basis: Rosenbaum & Pearl Ch. 1: adjustments to reported financial data.

Reference

What does the evidence support today?

Question for the analyst

The issue is **whether the supplied evidence supports treating the identified costs as non-recurring.**

Information available	What it establishes	What it does not establish
Adjustment schedule: EUR 2.4m (FY2024A), EUR 1.4m (FY2025A)	Management's proposed amount and description of the adjustment	Independent support for the amount; whether the costs are separable from ordinary operating expenses
Management states: duplicate-QA activity ended Nov-25; commissioning support ended Dec-25	Management identifies specific activities and states finite end dates	That the activities actually ceased; that related costs will not remain in the future operating cost base
No invoices or supporting schedules supplied	—	Independent verification of the amounts and the nature of the costs
No FY2026 staffing / cost plan or FY2026 budget supplied	—	Whether equivalent costs are expected to recur in continuing operations

What would strengthen the case?

Evidence that allows the analyst to verify:

- the amount of the identified costs;
- their separability from ordinary operations;
- that the underlying activities have ceased;
- whether equivalent costs remain in the expected future cost base.

Conclusion today

The available information supports the existence of a **management-proposed adjustment**. It does **not yet provide sufficient independent support**. Therefore: **preserve both EBITDA bases**.

How an EBITDA basis change propagates

Start from one change

$$\text{Basis B EBITDA} = \text{Basis A EBITDA} + \text{management-proposed adjustment}$$

If we preserve both bases consistently, that difference must propagate through every calculation that uses EBITDA or EBIT.

WHAT CHANGES

Metric	Why it changes
EBITDA	Basis B adds the adjustment.
EBITDA margin	EBITDA changes (revenue unchanged).
EBIT Net debt / EBITDA	$\text{EBIT}_{A,B} = \text{EBITDA}_{A,B} - \text{D\&A}$ Net debt is unchanged; the EBITDA denominator changes.
FCF under the stated simplified control	EBIT changes, so tax-effected EBIT changes.

WHAT DOES NOT CHANGE

- Revenue, D&A, Gross debt, Eligible cash, Net debt, CapEx
- Operating working capital, Change in operating working capital

FY2025A net debt-to-EBITDA: same numerator, different denominator

Start with the numerator

The EBITDA treatment does not change debt or cash:

$$\text{Net debt} = \text{Gross debt} - \text{Eligible cash} = 198.0 - 28.0 = \boxed{170.0}$$

Basis A: Net debt / reported EBITDA.

Reported EBITDA = 68.8

$$\hookrightarrow 170.0/68.8 = \boxed{2.47x}$$

Basis B: Net debt / adjusted EBITDA.

Management-adjusted EBITDA = $68.8 + 1.4 = 70.2$

$$\hookrightarrow \frac{170.0}{70.2} = \boxed{2.42x}$$

$$\boxed{2.42x - 2.47x = -0.05x}$$

$\hookrightarrow$ Basis B makes TargetCo appear modestly **less levered** than Basis A.

P3: preserve and compare both EBITDA bases

P3 changes the question

Do **not** ask AI which EBITDA basis is “best.” Ask it to show the financial consequences of **both supplied bases** while preserving the unresolved evidence question.

USE P3

Open the Session 2 prompt card and paste **P3 - Compare the EBITDA bases**.

The AI must preserve:

- 1) **Basis A:** reported EBITDA;
- 2) **Basis B:** management-adjusted EBITDA.

For FY2024A/FY2025A, propagate both through:

- EBITDA; EBITDA margin;
- EBIT; net debt-to-EBITDA;
- FCF under the stated simplified control.

REQUIRED OUTPUT

For each metric, require:

- Basis A; Basis B; Difference; Why it changes;
- Resolvable vs. Evidence needed.

Two prohibitions

The AI must not:

1. select one EBITDA basis as the approved treatment;
2. invent a partial adjustment between the two.

Consistency test

Trace: Basis A inputs/outputs; Basis B inputs/outputs.

A row is inconsistent if one component uses Basis A while another uses Basis B without an explicit analytical reason.

The purpose of P3 is comparison, not selection.

Student work: complete and verify the A/B comparison

WORK IN PAIRS — 8 MINUTES

Use the P3 output as a **draft**. Your deliverable is the independently verified table in 05_EBITDA_Bases.

1. COMPLETE THE TABLE

In the blank work area in 05_EBITDA_Bases:

1. enter the formulas for **Basis A** and **Basis B**;
2. complete both FY2024A and FY2025A for: EBITDA; EBIT;... FCF.
3. calculate: Difference = Basis B – Basis A.
4. keep the EBITDA basis explicit in every formula and label.

2. VERIFY WHAT AI PRODUCED

For every row:

1. trace the formula back to the historical workbook;
2. confirm that Basis A uses reported EBITDA consistently;
3. confirm that Basis B uses management-adjusted EBITDA consistently;
4. check the calculated difference independently;
5. complete *Resolvable today?* from the evidence available;
6. state the specific *Evidence needed* if the treatment remains unresolved.

Answer the questions:

What changes?

Why does it change?

Can the underlying treatment be resolved today?

Apply the verified corrections

Correct the objective errors

Use the corrections from the P2 debrief.

1. REPAIR THE WORKBOOK

Work from a fresh TargetCo copy.

1. Apply the corrections identified in the **P2 debrief**.
2. Rerun the relevant independent check:
 - revenue growth;
 - reported EBITDA margin;
 - EBIT dependency;
 - operating working capital;
 - net debt;
 - net debt-to-EBITDA basis.
3. Extra: for the repaired EBIT dependency, repeat the **perturbation test**.

2. DOCUMENT THE CHANGE

In 06_Audit_Log, record each underlying correction:

Field	Record
Location	Exact workbook cell or range
Before	Original formula or constant
After	Corrected formula
Reason	Verified construction error
Verification	Independent check performed
Result	Pass / Fail after correction

Do not change

- exceptions; blank; the reported / management-adjusted basis distinction.

Final control

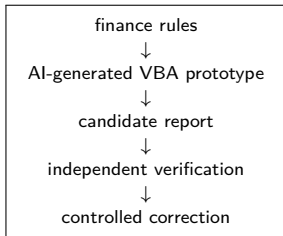
The repaired workbook should satisfy the objective construction checks **without concealing the unresolved EBITDA judgement**.

What leaves Session 2

Two outputs leave this session

The objective construction errors have been corrected and verified. The unresolved EBITDA treatment has **not** been converted into a false certainty.

1. A REUSABLE AUDIT WORKFLOW



2. A CONTROLLED HANDOFF TO VALUATION

Before Session 3:

- save the corrected historical workbook; complete 06_Audit_Log;
- retain both historical EBITDA bases; record the evidence still required to assess management's adjustment.

In Session 3:

- show the valuation effect of the reported and management-adjusted EBITDA bases;
- keep the EBITDA denominator explicitly identified;
- do not silently convert management's proposal into an analyst-approved normalization.

Different treatment for different types of uncertainty

Verified construction error	→	correct + test + document
Unresolved finance judgement	→	preserve + compare + carry forward

Appendix: six-sheet workbook

01_Raw_Data	Supplied facts and definition notes
02_Historical_Analysis	Historical analysis and cash-flow control
03_Adjustments	Management bridge and open evidence controls
04_Summary	KPIs, leverage and independent checks
05_EBITDA_Bases	Blank student work areas for the two-basis comparison
06_Audit_Log	Changes, exceptions and unresolved decisions
